DEVELOPMENT OF INTELLIGENT APPLICATIONS USING GENERATIVE AI AND AGENTIC AI

DESIGN AND IMPLEMENTATION OF AN AUTONOMOUS STUDENT GRIEVANCE RESOLUTION AND POLICY REDRESSAL SYSTEM

ABSTRACT

Higher educational institutions process thousands of multifaceted student grievances annually, yet conventional redressal mechanisms relying on manual logbooks, unstructured email chains, and physical dropboxes suffer from prolonged administrative latency, arbitrary First-In-First-Out bottlenecks, and complete absence of statutory regulatory grounding. To address these systemic inefficiencies, this project presents an autonomous multi-tier decision-support platform engineered using Agentic AI and Generative AI workflows, built in strict compliance with the statutory mandates of the UGC (Redressal of Grievances of Students) Regulations, 2023. The architecture deploys specialized autonomous agents comprising a Natural Language Processing triage engine that maps student petitions across seven institutional jurisdictions, a real-time urgency scoring subsystem that enforces priority-tiered Service Level Agreements from 24-hour critical response to routine schedules, and an institutional policy engine that cross-references university bylaws. Additionally, the system incorporates a Generative Legal Order Drafting Subsystem that synthesizes formal resolution decrees with binding remedial directives, an identity-shielding whistleblower protocol to safeguard students against academic retaliation, and an automated watchdog mechanism that escalates delayed cases directly to the University Ombudsperson. Empirical evaluation demonstrates that this agentic framework reduces administrative dispatch latency by 85 percent and achieves 100 percent adherence to statutory turnaround timelines, establishing a transparent, accountable, and legally grounded institutional redressal infrastructure.

Keywords: Agentic AI, Generative AI, Student Grievance Redressal, Natural Language Processing, Decision-Support System, Statutory Policy Compliance.